

nally conceived as educational tools. Several factors mitigate in favour of these platforms, including:

- These are affordable and compact.
- Their popularity means there is plenty of code in the open source community to help get basic features running.

So far so good, but caution is needed when taking the project forward. **Engineers should always consider how their chosen hardware will perform in the target environment**, including whether it is specified over an appropriate temperature range, or if physical interconnects will withstand expected levels of shock, vibration or moisture ingress.

If the chosen module is not correctly specified, extra costs may be incurred to carry out additional tests of design-in extra protection for hardware.

Other issues to consider are how the resulting product will perform in electromagnetic compatibility (EMC) tests. Although, the chosen board may have passed relevant EMC tests to qualify for CE marking, original equipment manufacturing (OEM) teams need to ensure the finished product will meet the requirements of their target market.

One example is a system that contains a projected-capacitive (PCAP) touchscreen user interface. If the touchscreen controller is an ordinary commercial-grade IC, it can be particularly difficult to satisfy IEC 61000-4-6 Level 2 or 3 conducted noise immunity specifications for industrial or medical equipment. A higher-grade controller may be required, and changes to the board layout may also be needed to achieve the mandatory immunity.

It is also critical to ensure that the module—as a constituent component—will be both adequately supported and available to purchase for the intended lifetime of the end product. This is a given for industrial components, which are typically backed up by a manufacturer's longevity commitment for as many as ten years.

Consumer components, on the other hand, come with no such assurances, and the desired sup-

port commitment from original module manufacturers may not be provided, and could result in costly re-designs halfway through the project's lifetime.

Moreover, module producers sometimes have to make small changes to their products that can affect such parameters as signal timing or voltage thresholds. Engineers need to know how their system's performance may be affected. While, a vendor of industrial-grade products will likely provide dedicated support to answer such questions from customers, the same is not so easily provided by an educational board producer.

Industrialise early

Encountering these issues later in the project, when extra effort is needed to work around them, can prevent reaching critical milestones on time. Having to change hardware to overcome these barriers can introduce delays and expenses that may negate the expected advantages of using open source hardware.

Porting the design to an industrial-ready platform at an early stage can be easier than expected. An HMI design expert can help ensure a smooth and efficient transition from the initial proof-of-concept model to a more industrialised and rugged prototype suitable for subsequent stages of development.

To establish a suitable hardware platform for development, a strong choice of application processor is a good starting point. While it is true that some chip makers are focused mainly on the consumer space, where short lifecycles prevent longevity assurances, several manufacturers offer products that are suitable for more specialised industrial or medical applications.

One example is NXP i.MX6 family. These are available in single-, dual- or quad-core versions and feature a combination of Arm Cortex-A9, A7. These support high-performance interfaces that are popular in industrial applications, such as SATA 2, FlexCAN, PCIe and Media Local Bus (MLB) support for easy connection to Media Oriented Systems Transport (MOST) or other

networks, MIPI display and camera ports.

These features are in addition to well-known connectivity standards such as Ethernet up to 1Gbps, USB Host and multiple ADC channels. In addition to multi-core scalability, this family also offers flexible graphics options with 3D support and dedicated acceleration engines that outperform the general-purpose graphics support integrated in processors of low-cost educational platforms.

Multiple variants offer users a wide variety of configurations including some models with a built-in Cortex-M4 co-processor to offload the main application processor. This maximises flexibility to meet system performance requirements and budgetary constraints. At the entry level, pricing can compete favourably with open source alternatives.

HMI design experts are familiar with this and other families of application processors that provide a suitable foundation for industrial-ready HMI control. They can help establish a suitable hardware platform based on any of these devices, or quickly port a proof-of-concept model to a platform that can more readily be prepared for industrial use-cases.

Conclusion

The opportunity to trim time and cost from a development project by starting with affordable off-the-shelf hardware can be tempting, especially for designers who are familiar with the finer points of such platforms as Raspberry Pi and Arduino. Although, these platforms can be effective for proof-of-concept work, problems may arise when industrialising the end product to withstand the operating environment and satisfy mandatory requirements of the target market.

If these criteria cannot be met, **a recommended approach is to port the design to a more robust, industrial-oriented platform at an early stage of the project.** Ultimately, this may result in faster time-to-market and lower development costs. Seeking expert advice is always recommended, to help commit to the best hardware for the intended application at the right time in the project. **EFY**